

Coupon Coin (CPC)

Capstone Final Report — Figure Atlas

29 figures · auto-generated · vector PDF

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Global coupon redemption rates by industry segment (2024)

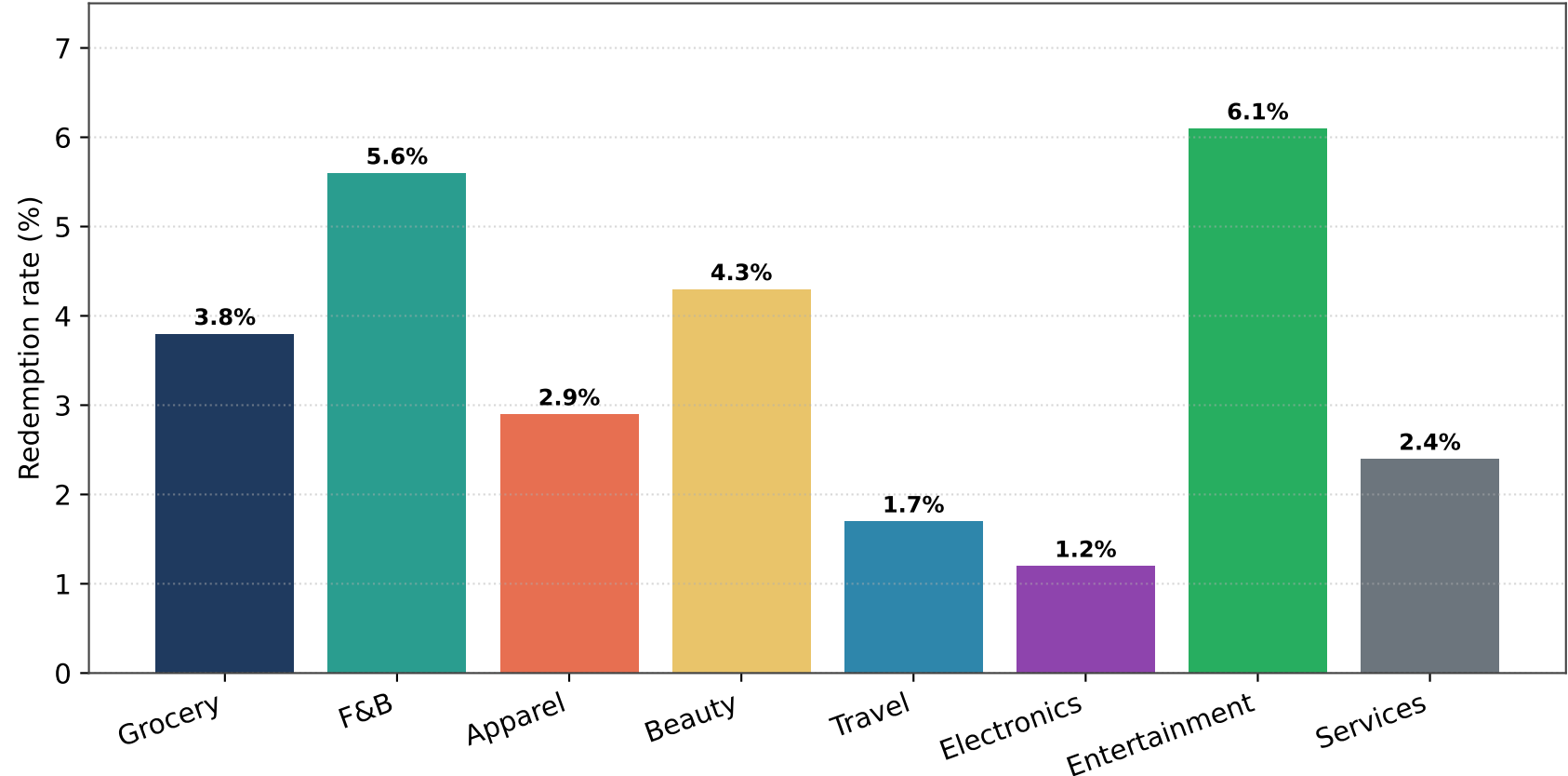


Figure 1. Global coupon redemption rates by industry segment (2024)

Comparison of traditional vs blockchain-based coupon systems

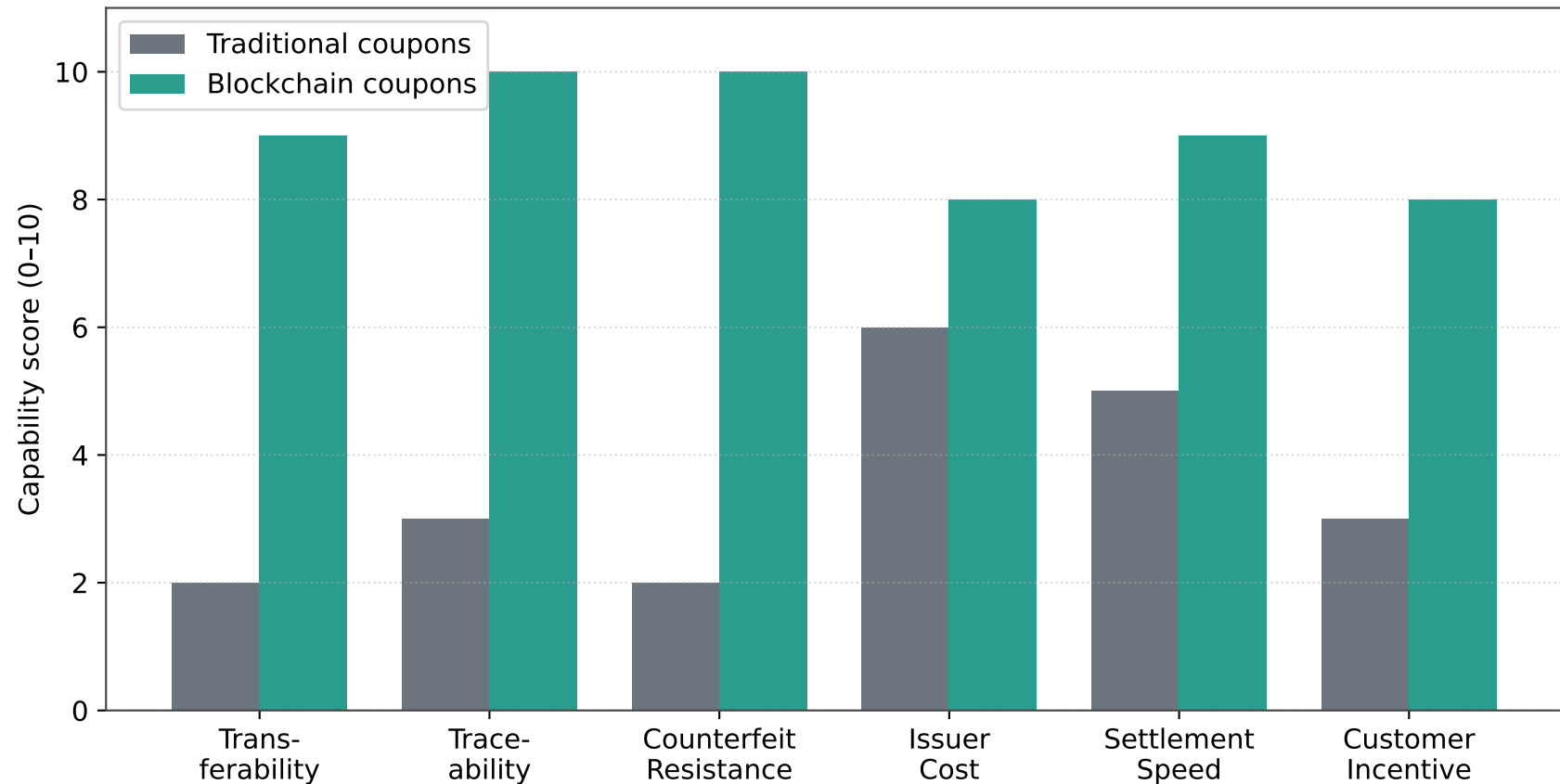


Figure 2. Comparison of traditional vs blockchain-based coupon systems

Ecoupon-Chain Decentralised Coupon Architecture

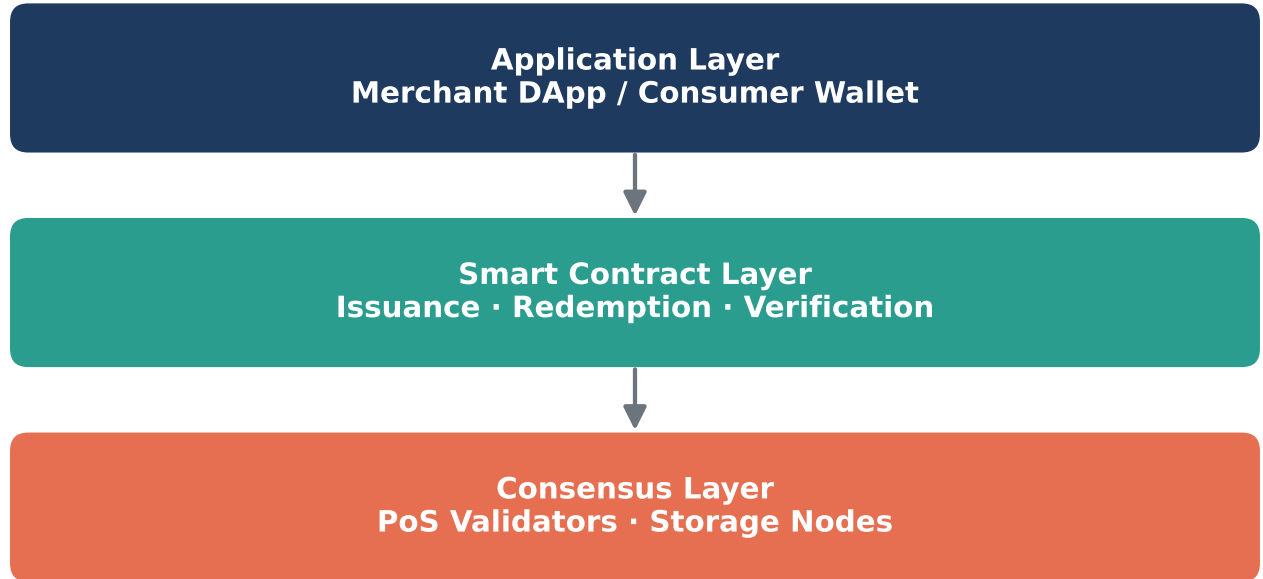
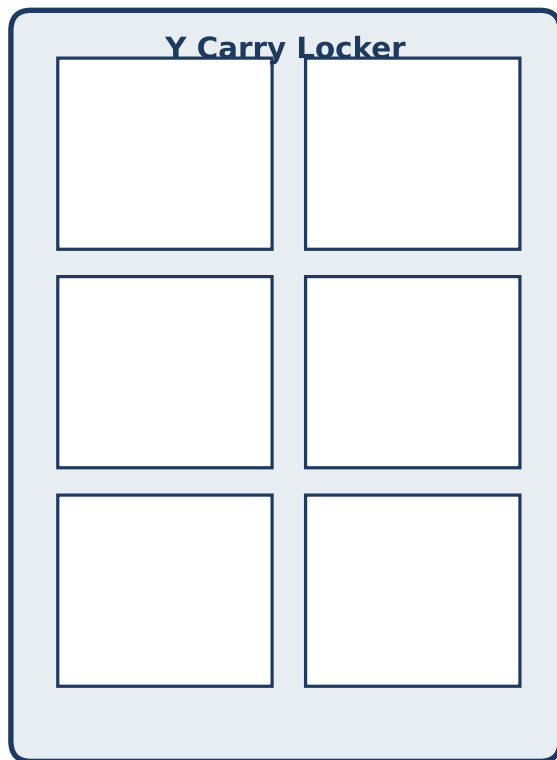


Figure 3. Ecoupon-Chain decentralised coupon architecture

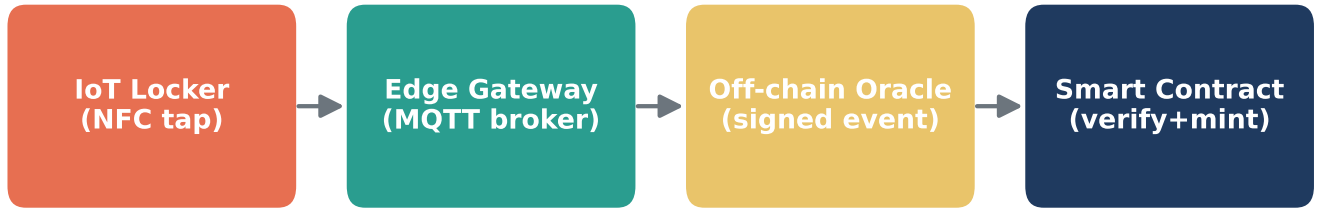


Hardware Specification

- **Chassis:** Powder-coated 1.5 mm steel, IP54
- **Compartments:** 6 / 12 / 18 modular SKUs
- **Lock:** Solenoid bolt, 12 V DC, 800 mA
- **MCU:** ESP32-S3, dual-core 240 MHz
- **Connectivity:** Wi-Fi 802.11n + LTE Cat-M1
- **Reader:** NFC Type-A/B (PN532) + QR camera
- **Display:** 3.5" capacitive touch, 480×320
- **Power:** 100–240 V AC, UPS 24 Wh
- **Telemetry:** Temp / door / shock sensors
- **Cert.:** CE, FCC, RoHS

Figure 4. Y Carry Smart Locker hardware specification

Smart Locker Blockchain Integration Model



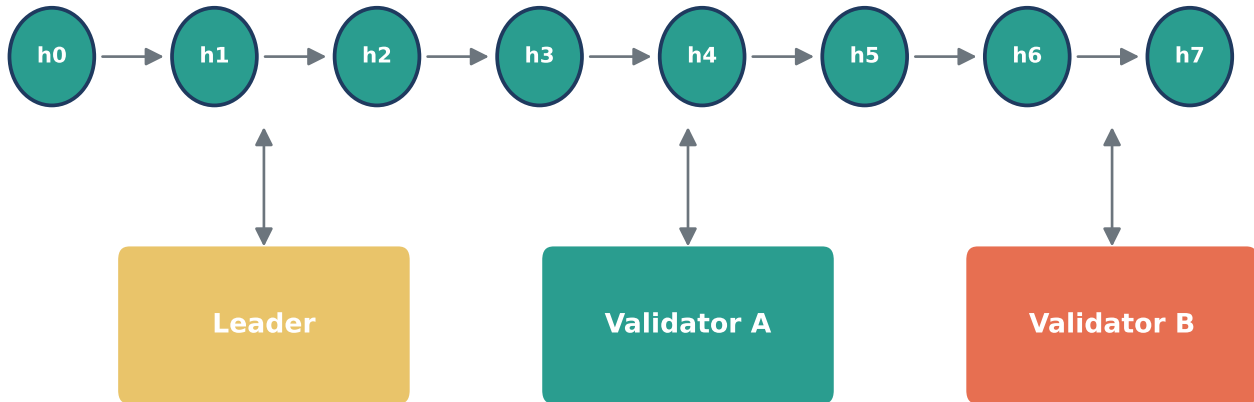
Hardware → Edge → Oracle → Blockchain (cryptographically attested)

Figure 5. Smart locker blockchain integration model

Solana Proof-of-History Consensus Mechanism

Verifiable Delay Function (PoH)

$$\text{hash}_n = \text{SHA-256}(\text{hash}_{n-1} \parallel \text{data}_n)$$



Tower BFT consensus over PoH timeline · ~400 ms slot · 65 k TPS theoretical

Figure 6. Solana Proof-of-History consensus mechanism

Overview of the Y Carry Smart Locker Ecosystem

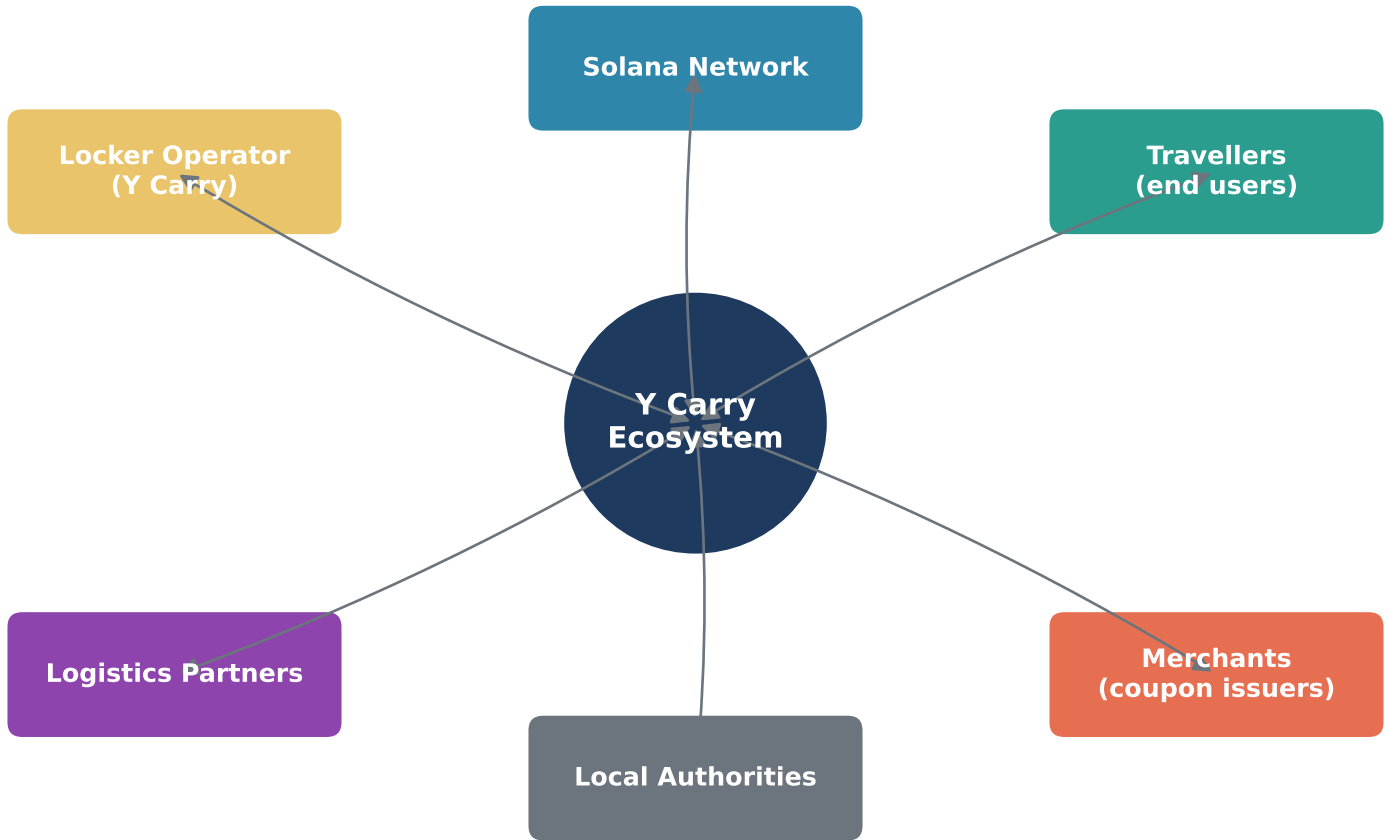


Figure 7. Overview of the Y Carry Smart Locker ecosystem

Y Carry Global Deployment (2025): bubble area $\propto$ locker count

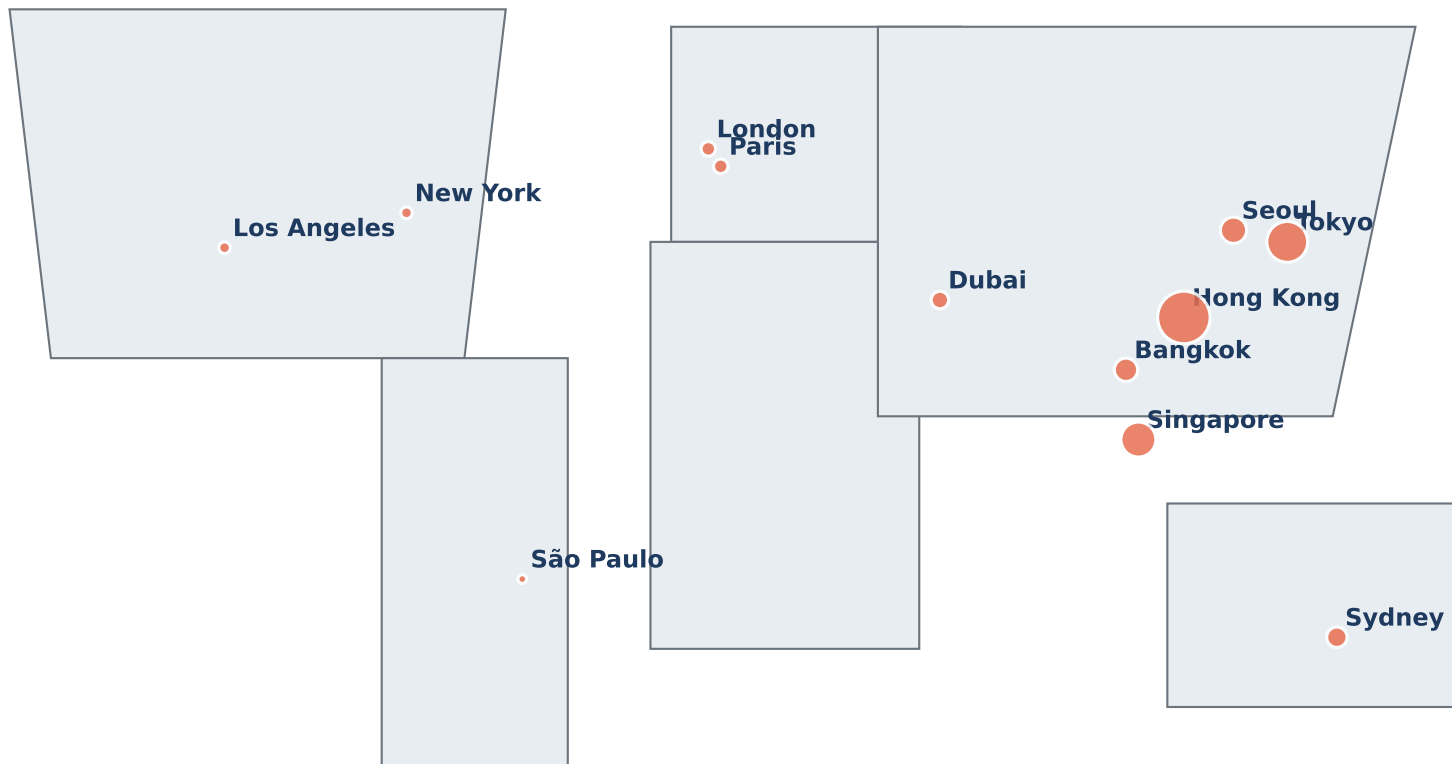


Figure 8. Y Carry global deployment map

CPC Protocol Three-Layer System Architecture

Presentation Layer

Mobile App · Web DApp · Merchant Portal · POS SDK



Protocol Layer

CPC Token · Locker Token · Reward Engine · Oracle



Settlement Layer

Solana SPL · L2 Rollup · Off-chain Indexer

Figure 9. CPC protocol three-layer system architecture

Blockchain Selection Radar — six candidate networks

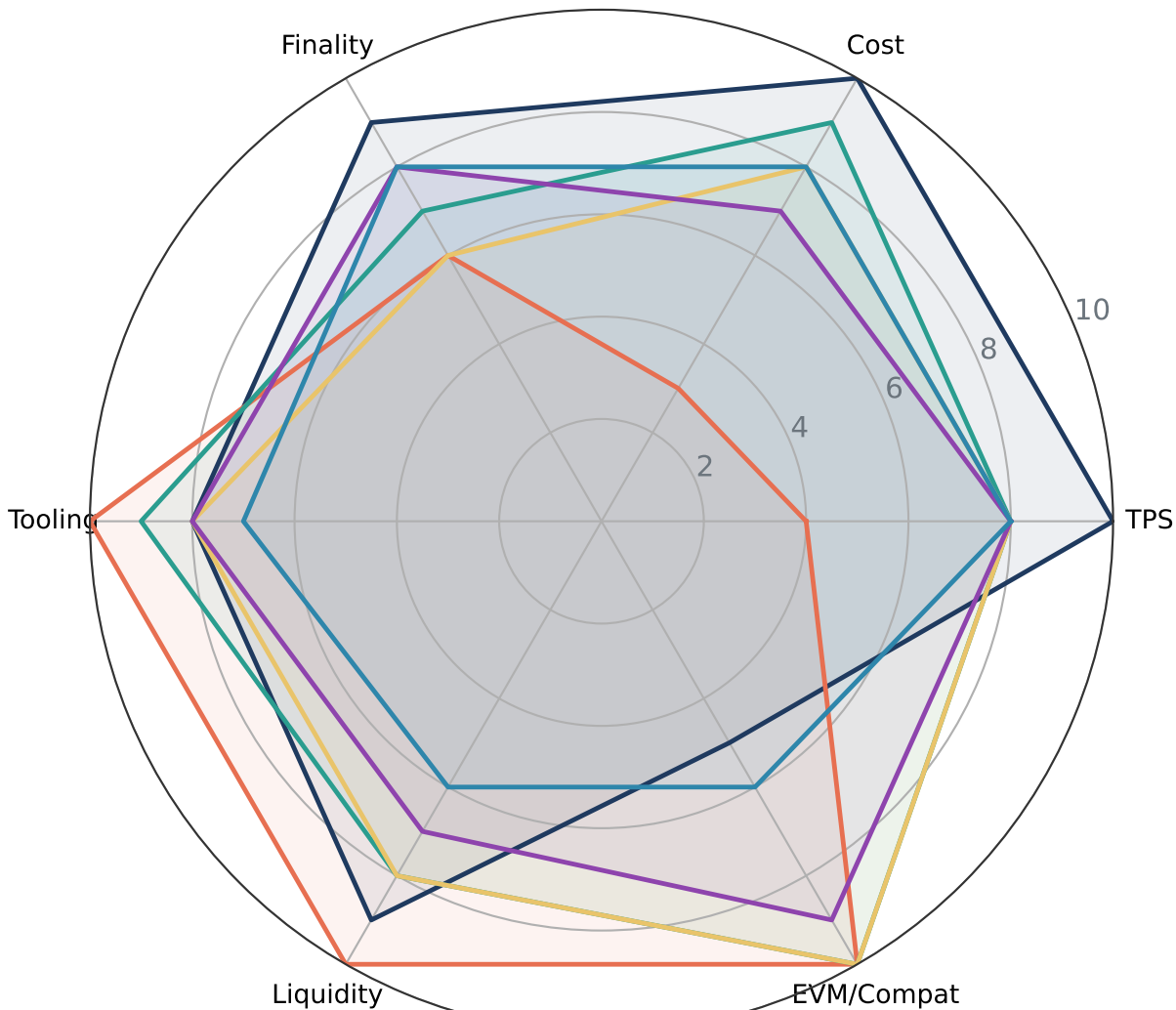
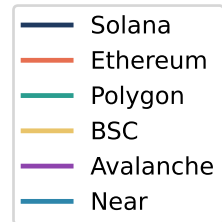


Figure 10. Blockchain selection radar chart across six candidate networks

CPC Token Distribution

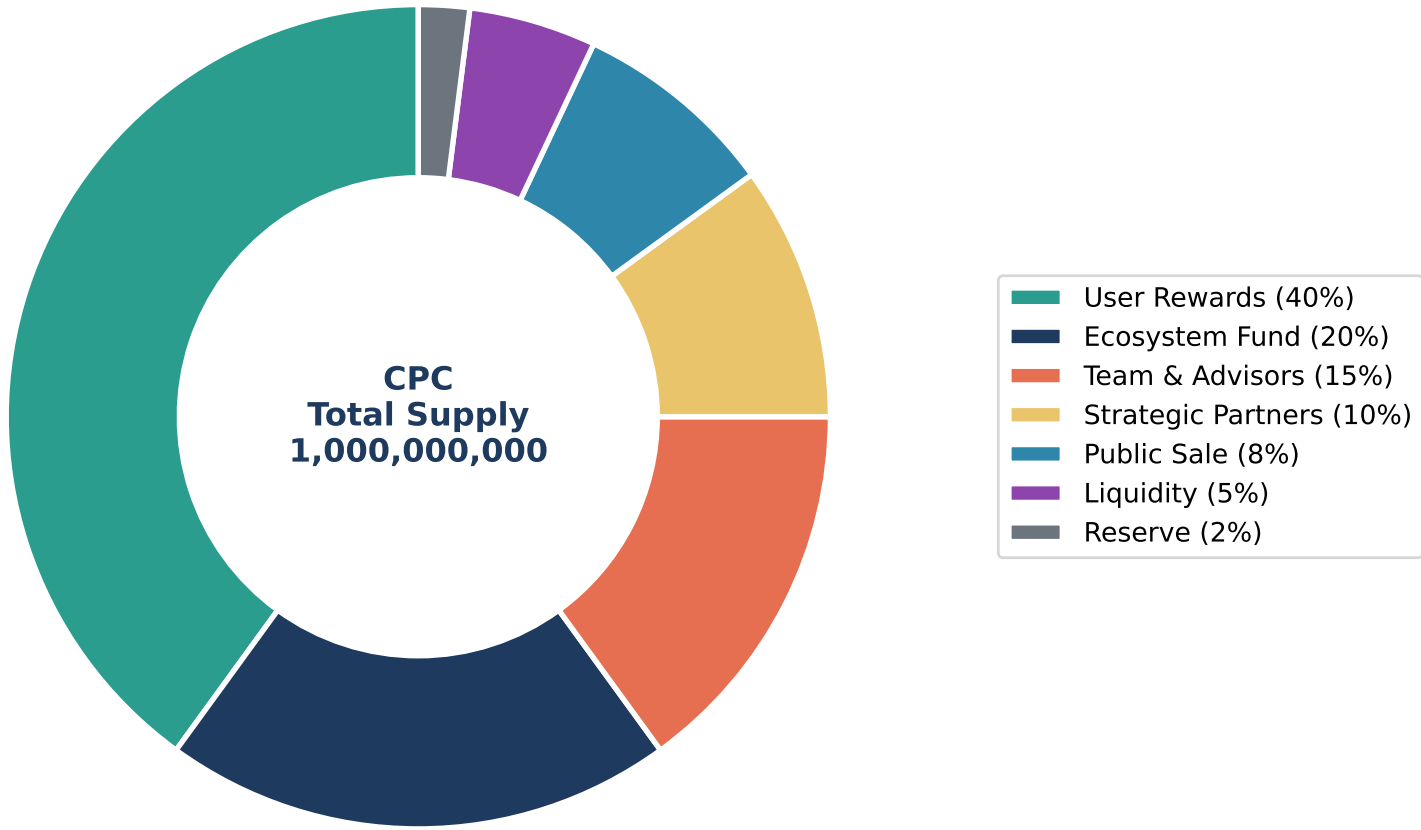


Figure 11. CPC token distribution doughnut chart

CPC Vesting Schedule — circulating supply unlock over 48 months

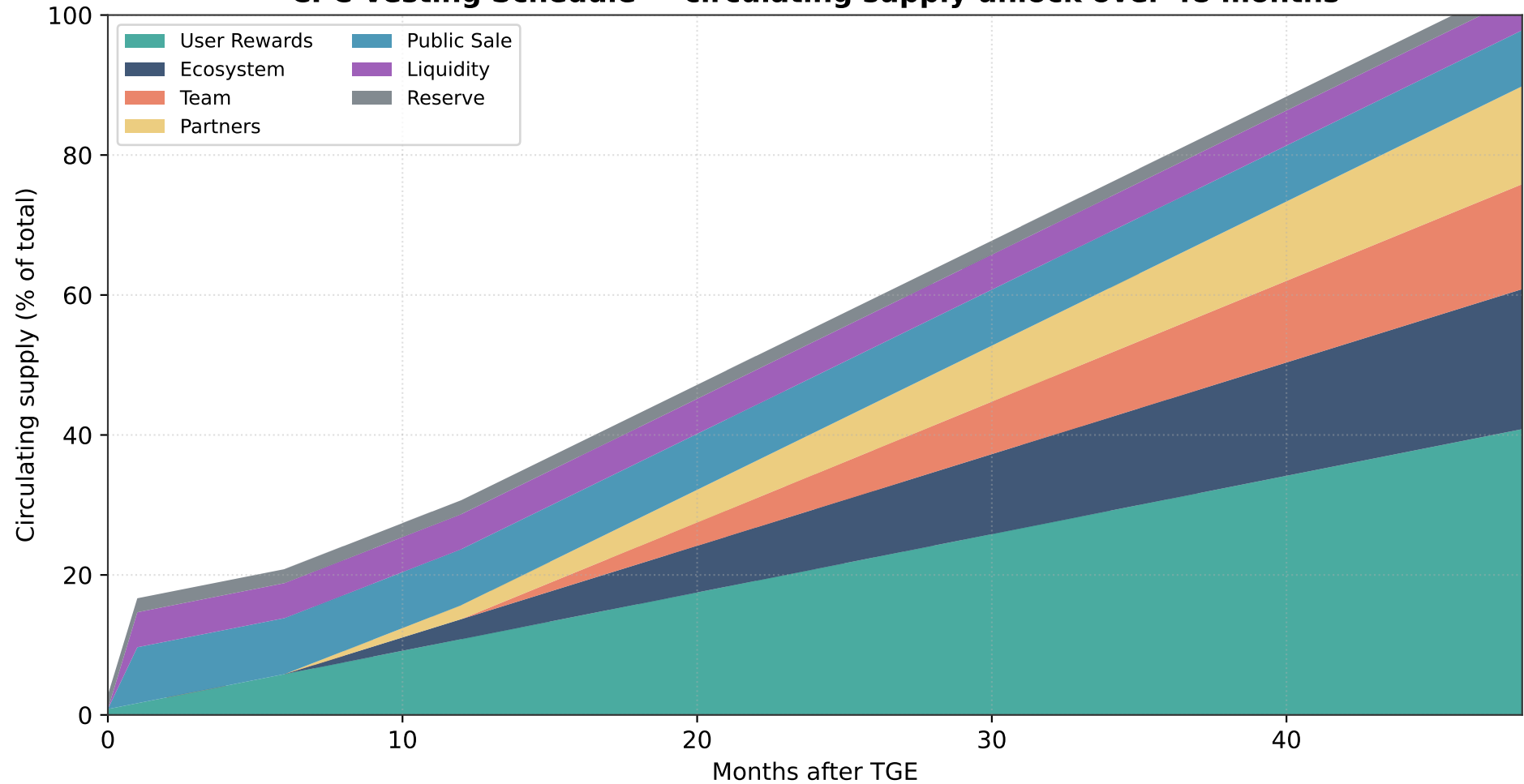


Figure 12. Token vesting schedule timeline

Deflationary Supply Simulation (5-year horizon)

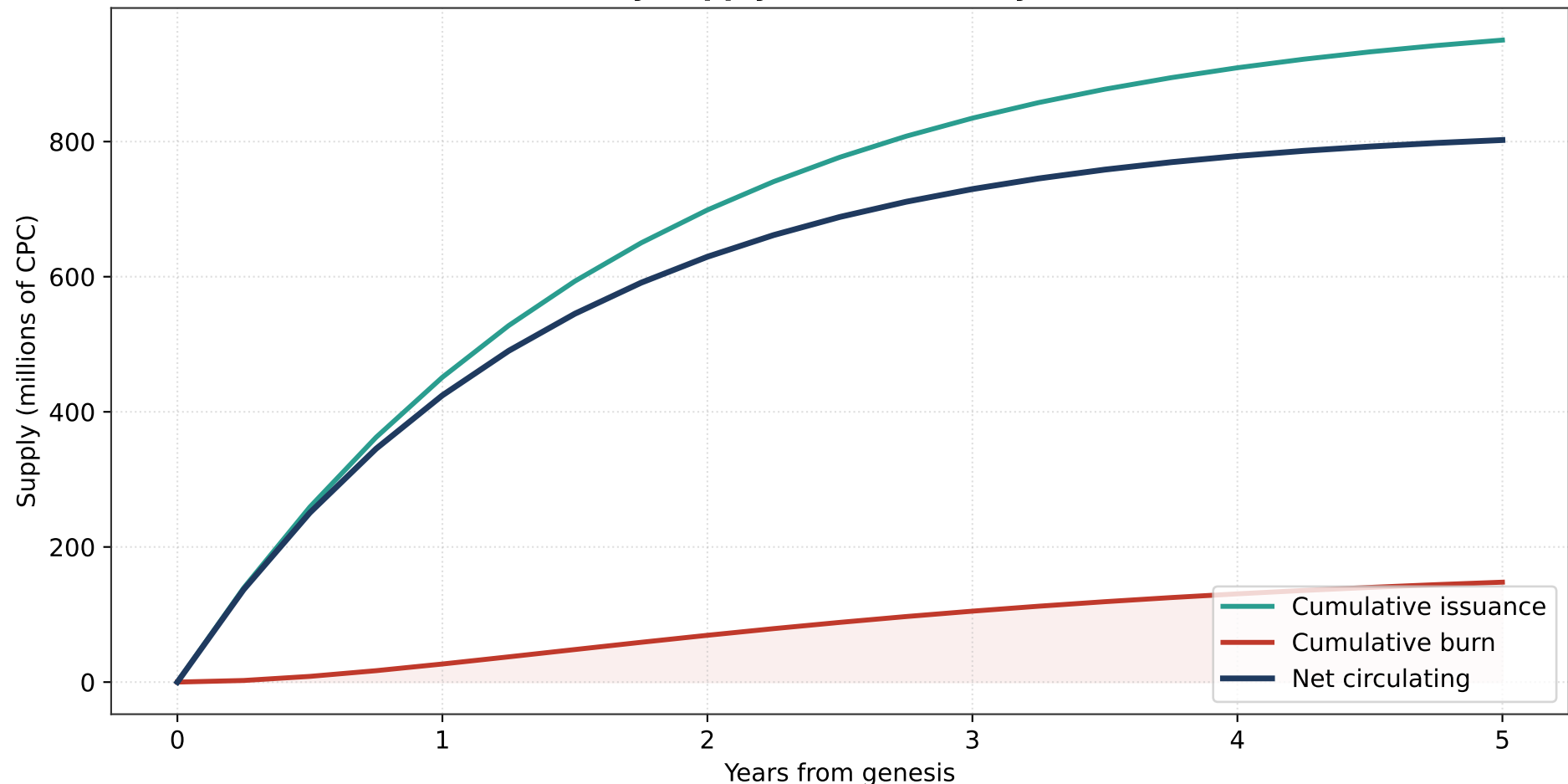
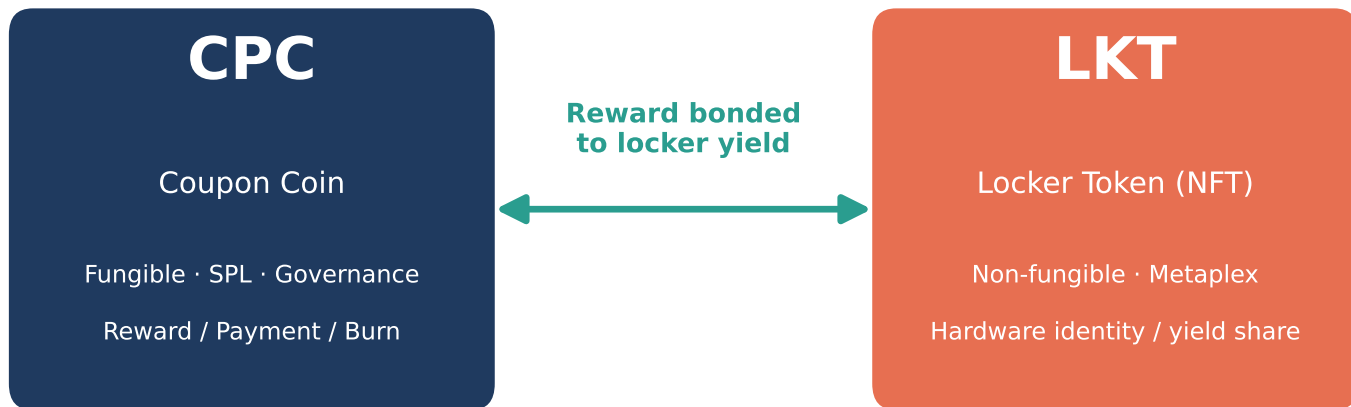


Figure 13. Deflationary supply simulation over five years

Dual-Token Architecture: CPC × Locker Token



CPC = utility/governance medium · LKT = real-world hardware credential

Figure 14. Dual-token architecture: CPC and Locker Token

Smart Contract Suite — interaction & dependency

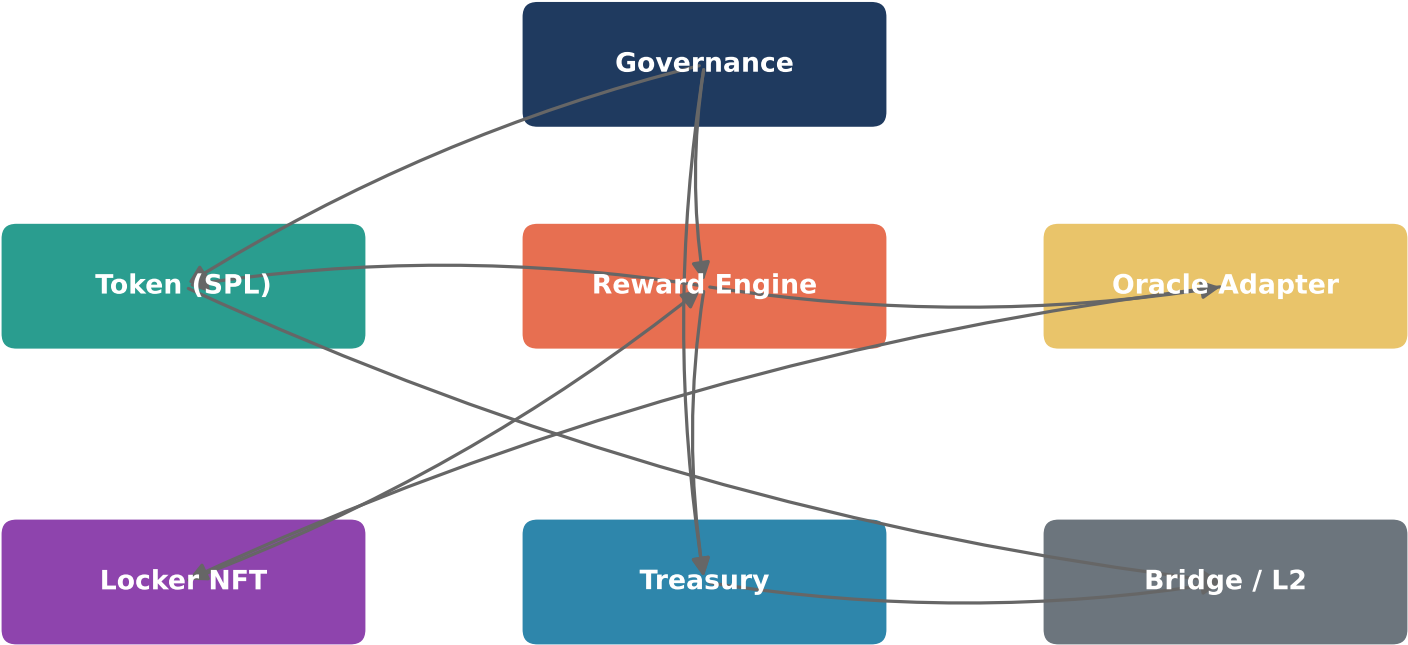


Figure 15. Smart contract suite interaction and dependency flow

Seven-Step Transaction Lifecycle (NFC tap → on-chain settlement)



Latency budget: ≤ 1.2 s end-to-end · finality ≈ 400 ms slot · cost $\approx \$0.00025$

Figure 16. Seven-step transaction lifecycle from NFC tap to reward distribution

Layer-2 Rollup: 30-second batching · Merkle settlement · 24 h challenge

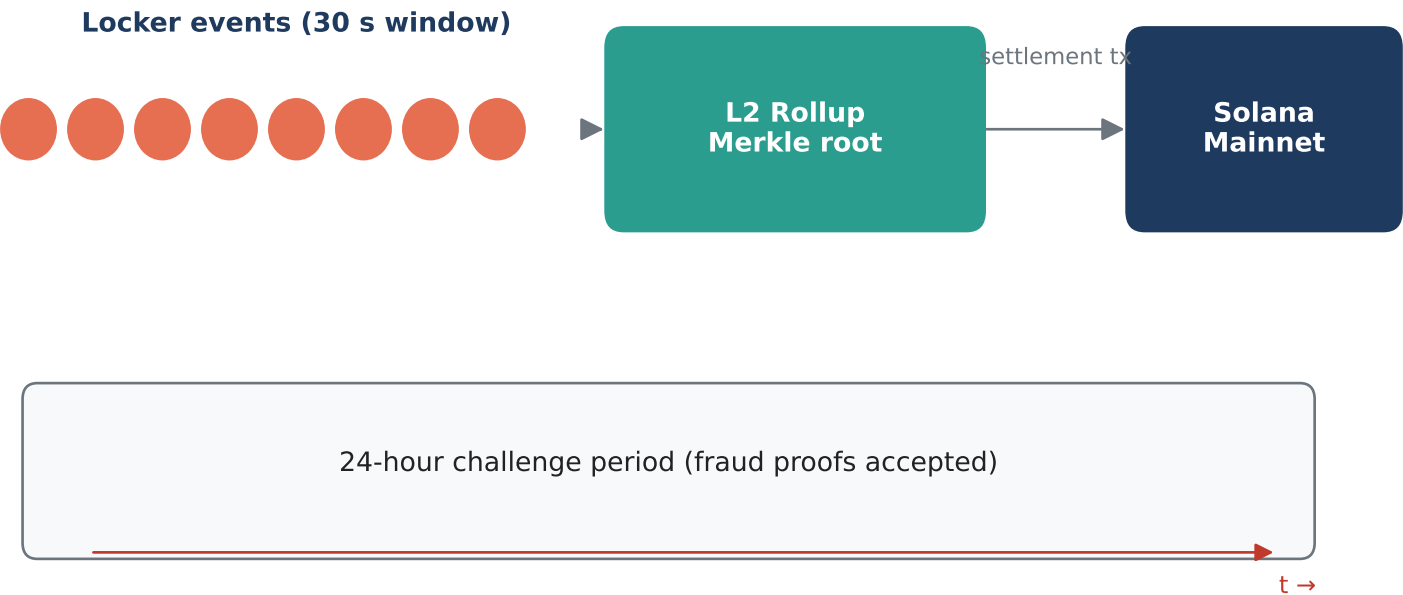


Figure 17. Layer-2 rollup batching and Merkle root settlement mechanism

```
$ solana config set --url https://api.devnet.solana.com
Config File: /Users/dev/.config/solana/cli/config.yml
RPC URL:      https://api.devnet.solana.com
WebSocket URL: wss://api.devnet.solana.com
Keypair Path: /Users/dev/.config/solana/id.json
$ spl-token create-token --decimals 6
Creating token EXjK9...rGyq
Address:  EXjK9bV2x3qP7vC4nU8tH2YwM1cVdRfL5pA8sBkQrGyq
Decimals: 6
Signature: 5Lq7e2bCdEf3HjK9...8mN4zVw2QpRsTuVwXyZ
$ spl-token create-account EXjK9...rGyq
Creating account 9pQ3...nM7T
Signature: 4xY2zT9eCm...8KjLwP3rNb
✓ CPC token deployed successfully on Solana Devnet
```

Figure 18. Solana Devnet: token deployment confirmation screenshot

[https] solscan.io/token/EXjK9...rGyq?cluster=devnet

Token · Coupon Coin (CPC)

Mint EXjK9bV2x3qP7vC4nU8tH2YwM1cVdRfL5pA8sBkQrGyq

Decimals
6

Total Supply
1,000,000,000 CPC

Holders
12

Type
Fungible (SPL)

Mint Authority
Hh3...9pK

Freeze Authority
—

✓ Verified on Solana Devnet · Last updated: 2026-04-21 14:32 UTC

Figure 19. Solscan Devnet: CPC Mint Address verification

zsh — CPC mint & balance check

```
$ spl-token mint EXjK9...rGyq 100000 9pQ3...nM7T
```

Minting 100000 tokens

Token: EXjK9bV2x3qP7vC4nU8tH2YwM1cVdRfL5pA8sBkQrGyq

Recipient: 9pQ3sD8fGhJkLmNpQrStUvWxYz1A2B3C4D5E6F7nM7T

Signature: 7Bc5...8aDpQ2eRfTgYhJk...wXyZ

```
$ spl-token balance EXjK9...rGyq
```

100000

```
$ spl-token accounts
```

Token

Balance

EXjK9bV2x3qP7vC4nU8tH2YwM1cVdRfL5pA8sBkQrGyq 100000

✓ Mint operation succeeded — balance verified

Figure 20. Token minting test: balance confirmation

Smart Contract Test Suite — 12/12 cases passing

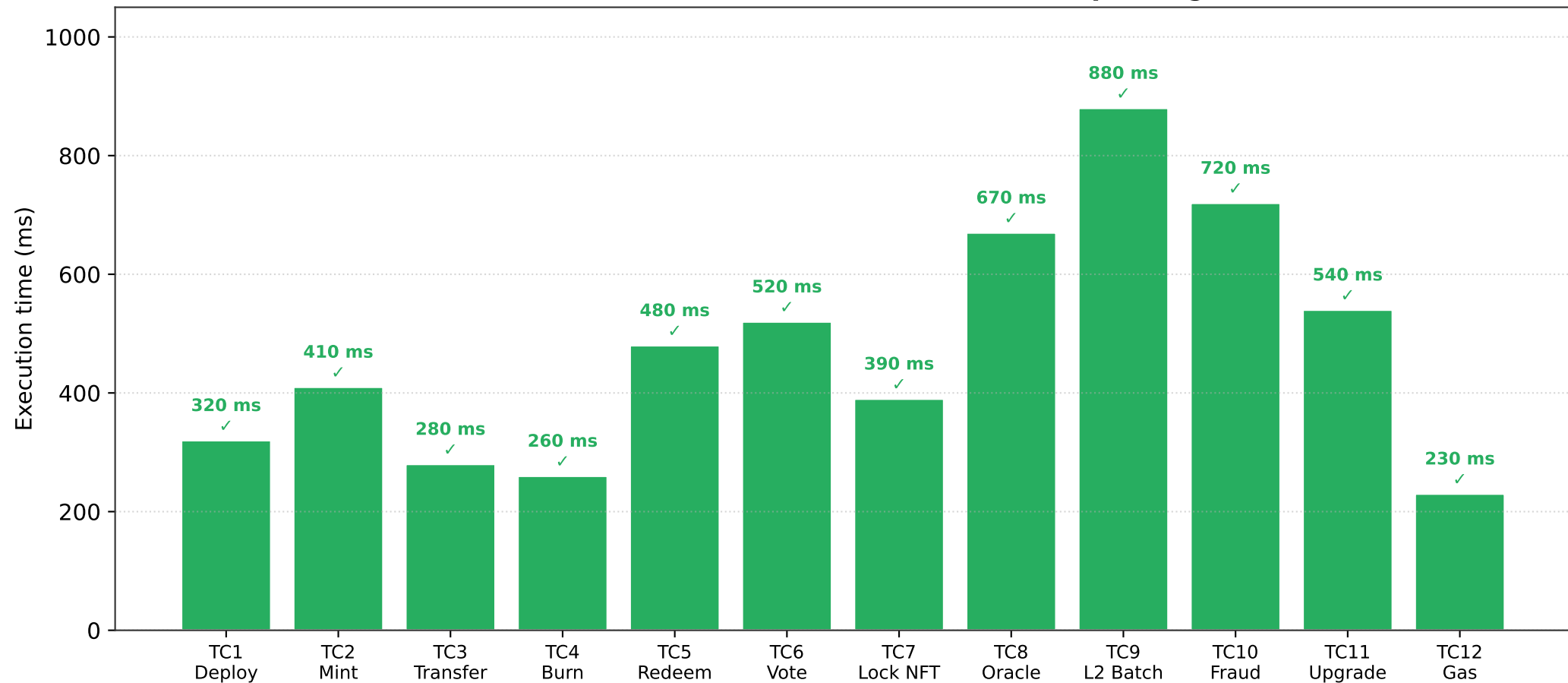
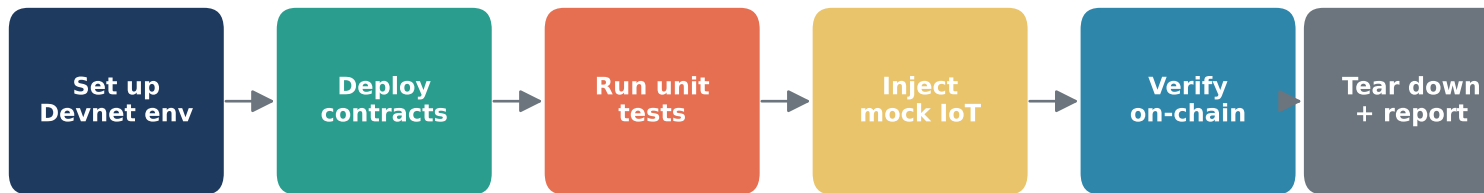


Figure 21. Smart contract test suite: results summary

Functional Test Flow — Devnet → Validation Pipeline



All 12 test cases produce JUnit XML; coverage $\geq 87\%$

Figure 22. Smart contract functional test flow

Transaction Lifecycle — sequence diagram

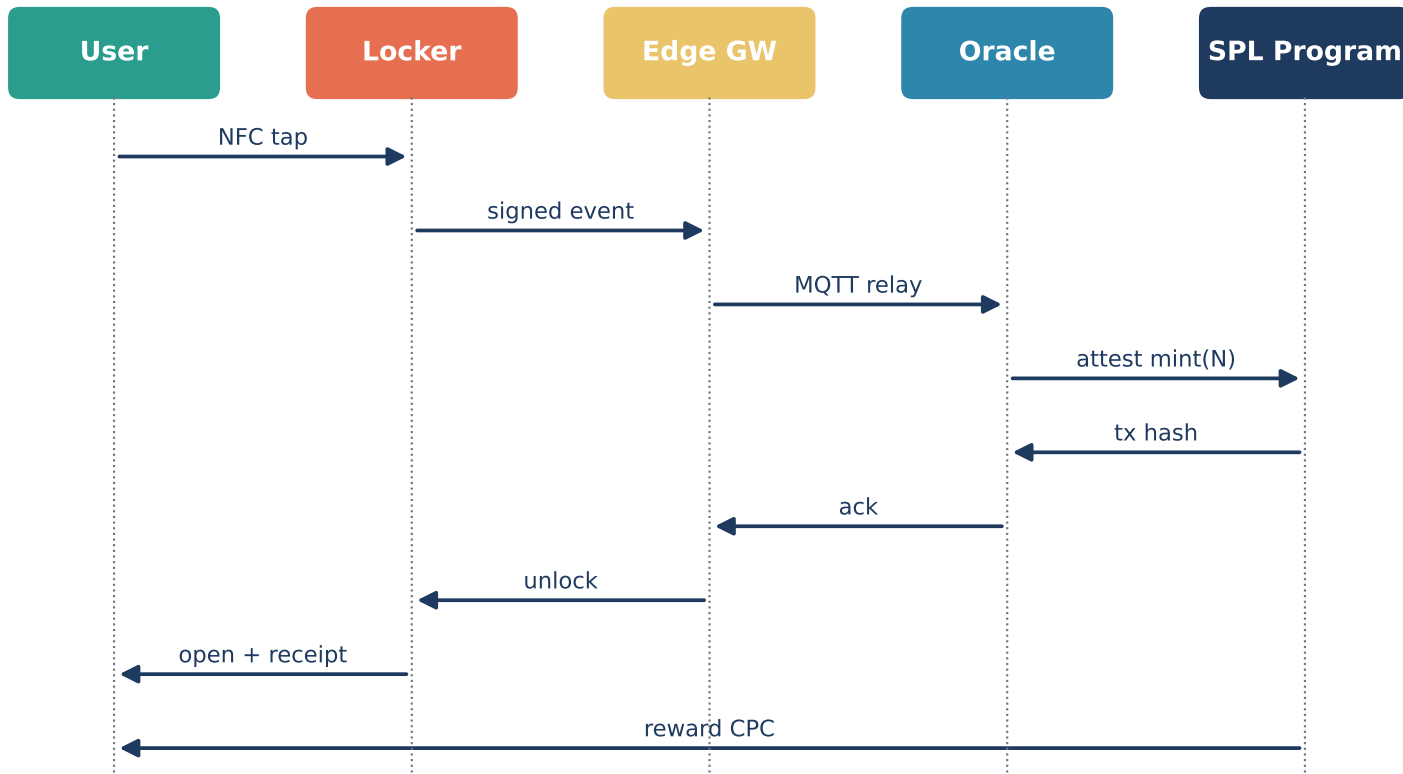


Figure 23. Transaction lifecycle: IoT event to on-chain settlement

Rollup Challenge Period & Finality Confirmation

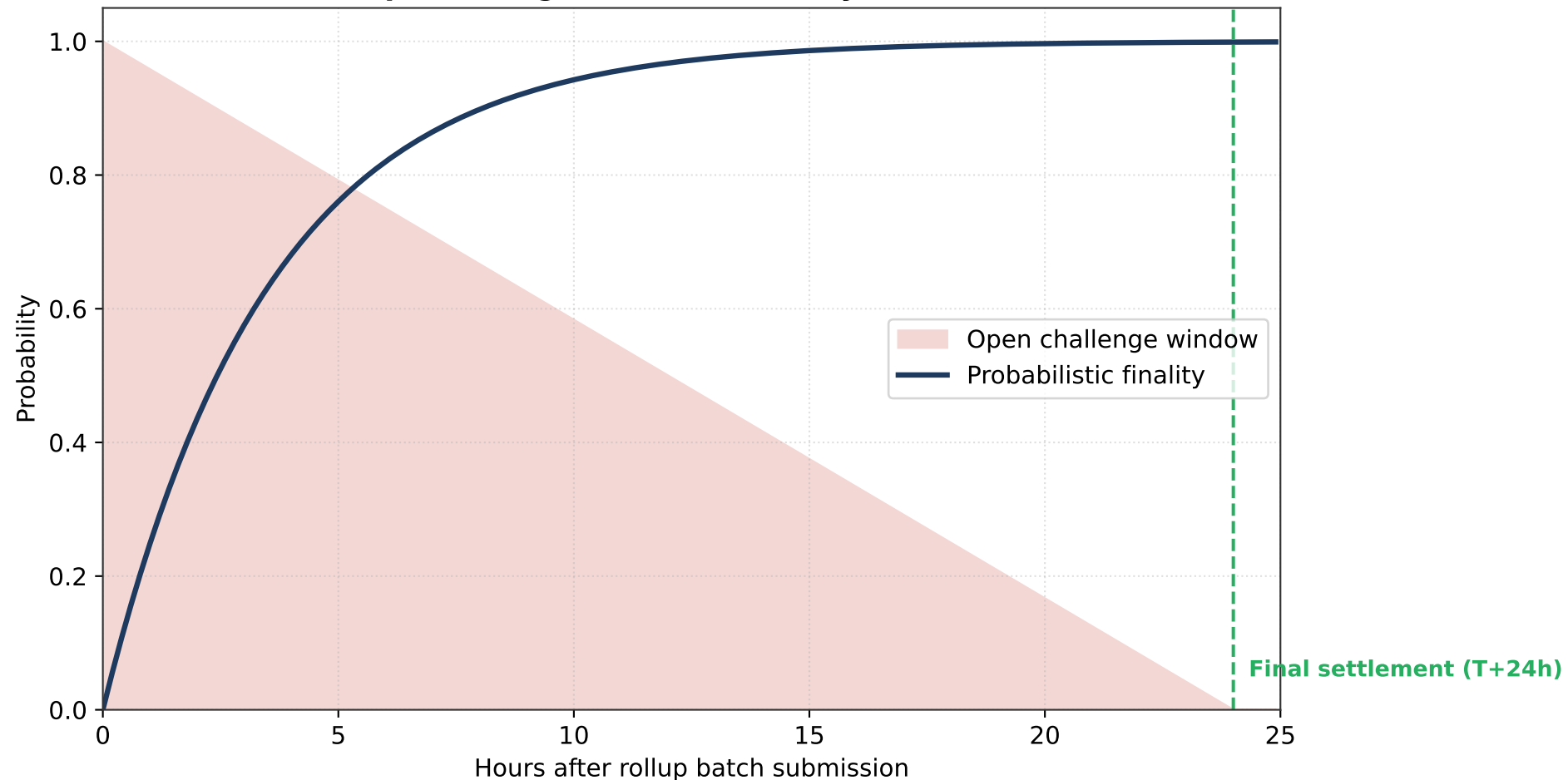


Figure 24. Rollup challenge period and finality confirmation

Three-Year Conservative Locker Deployment Projection

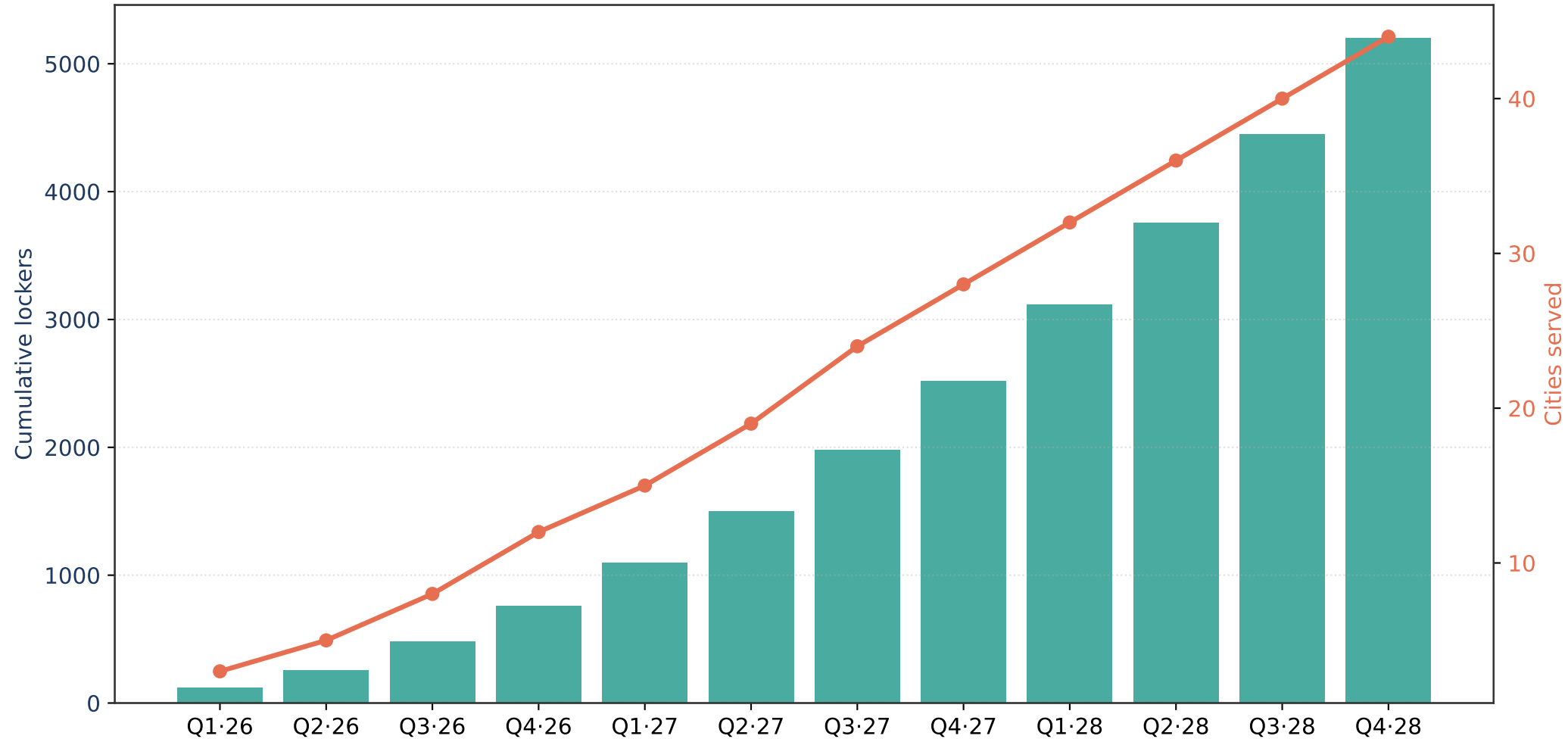


Figure 25. Three-year conservative locker deployment projection

CPC Revenue Model Flywheel

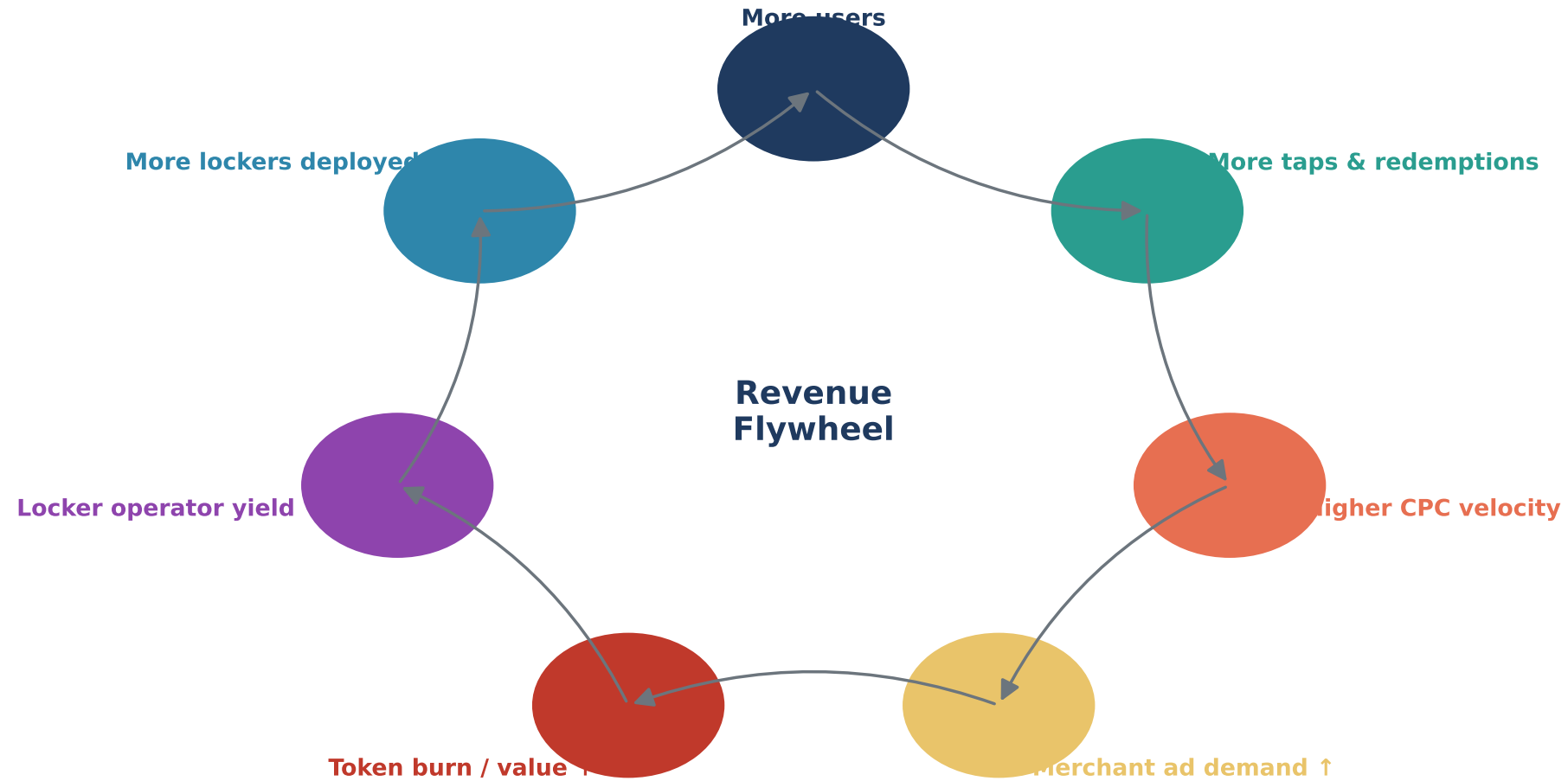


Figure 26. Revenue model flywheel

Risk Severity Matrix — CPC Protocol

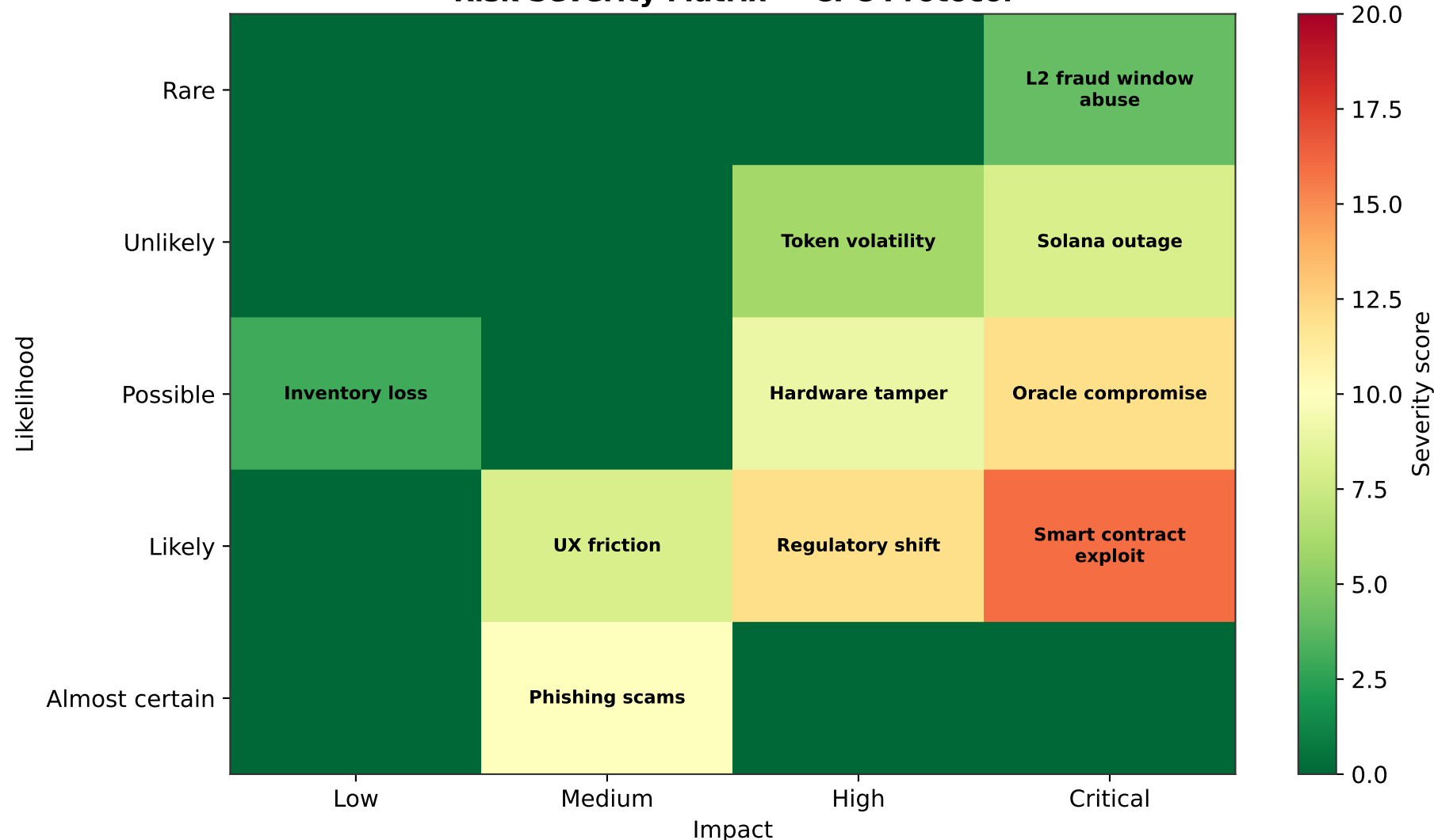
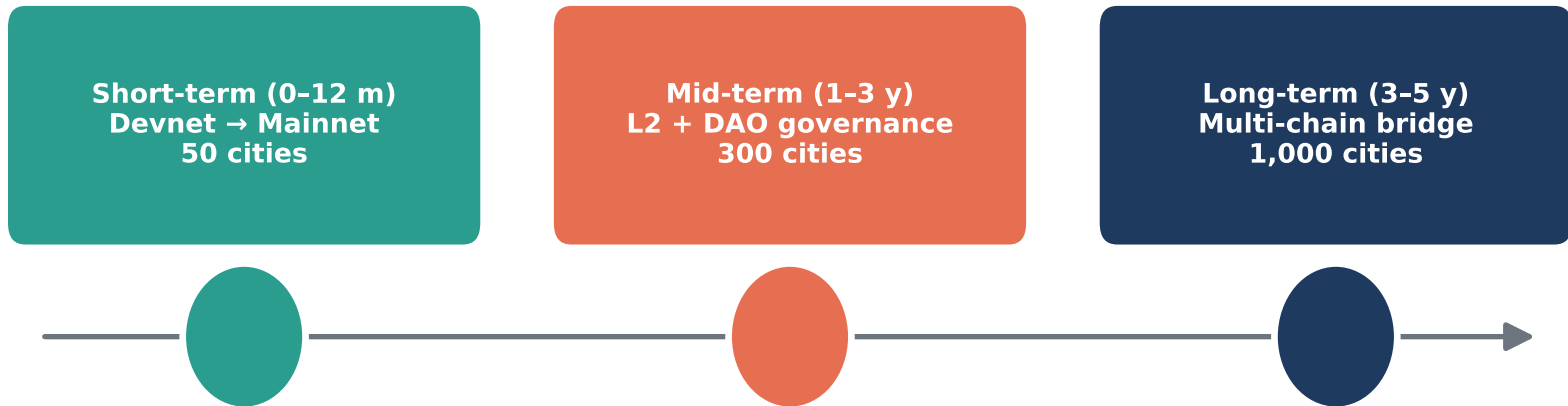


Figure 27. Risk severity matrix heatmap

Growth Trajectory — short-, mid- and long-term milestones

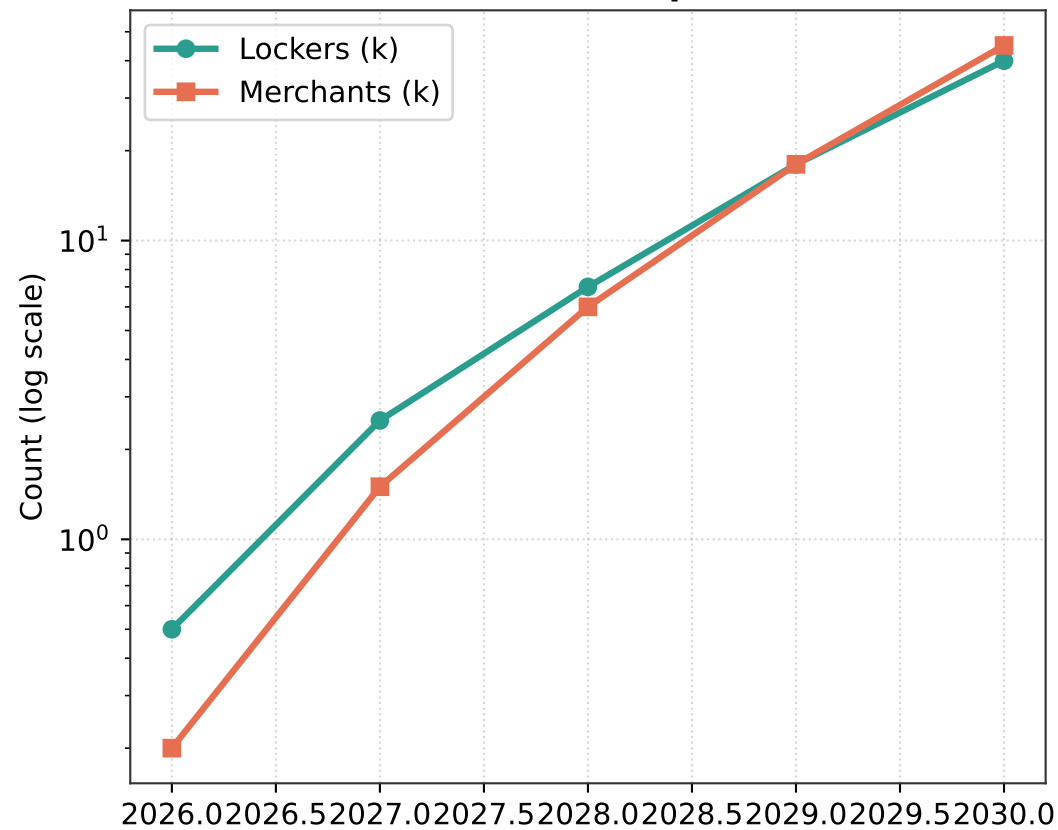


Milestones: TGE → Audit → L2 launch → DAO → Multi-chain

Figure 28. Growth trajectory roadmap: short-, mid-, and long-term milestones

Five-Year Ecosystem Scaling Vision (2026-2030)

Network footprint



Demand & monetisation

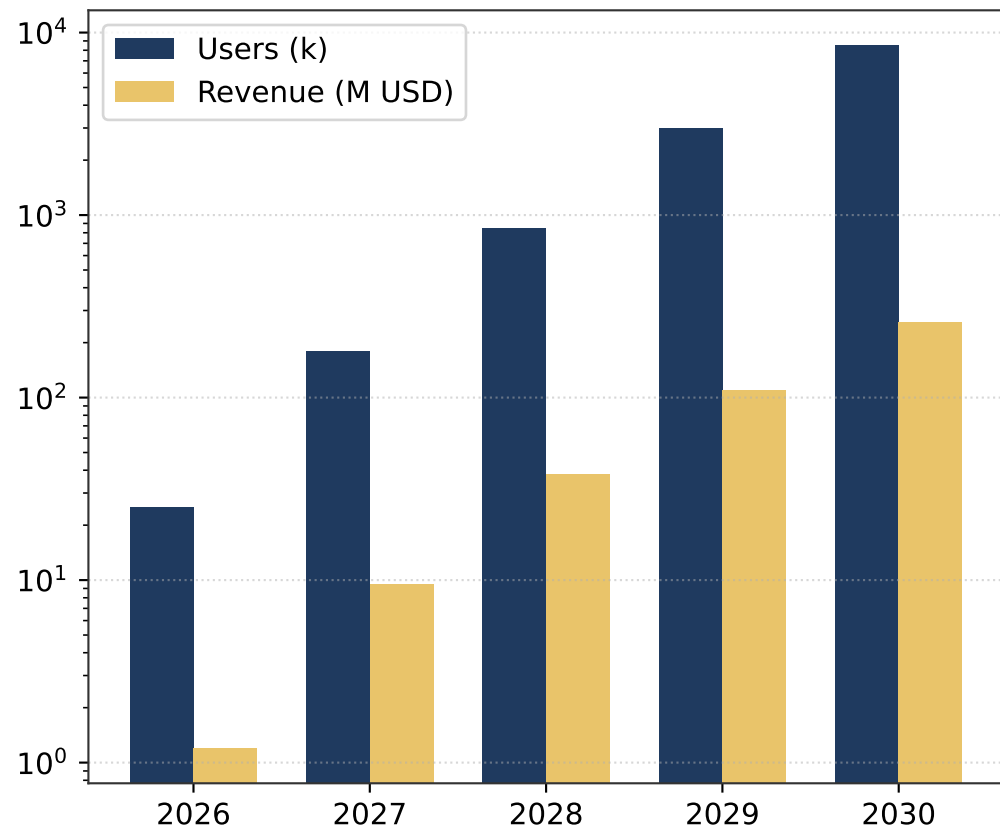


Figure 29. Five-year ecosystem scaling vision